



ER DIAGRAM DESIGN AND RELATIONAL SCHEMA

Assignment 2 : Hotel Management System

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OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Hotel Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ENTITIES AND ATTRIBUTES

Entity 1: GUEST

Column	Key
guest_id	Primary Key
guest_name	-
email	-
phone	-
address	-

Entity 2: ROOM_TYPE

Column	Key
room_type_id	Primary Key
type_name	-
price_per_night	-
description	-

Entity 3: ROOM

Column	Key
room_id	Primary Key
room_number	-
floor_number	-
room_status	-
room_type_id	Foreign Key

Entity 4: STAFF

Column	Key
staff_id	Primary Key
staff_name	-
role	-
phone	-
department	-

Entity 5: RESERVATION

Column	Key
reservation_id	Primary Key
guest_id	Foreign Key
room_id	Foreign Key
check_in_date	-
check_out_date	-
reservation_status	-

Entity 6: SERVICE

Column	Key
service_id	Primary Key
service_name	-
service_price	-
description	-

Entity 7: PAYMENT

Column	Key
payment_id	Primary Key
reservation_id	Foreign Key
payment_date	-
amount	-
payment_method	-
payment_status	-

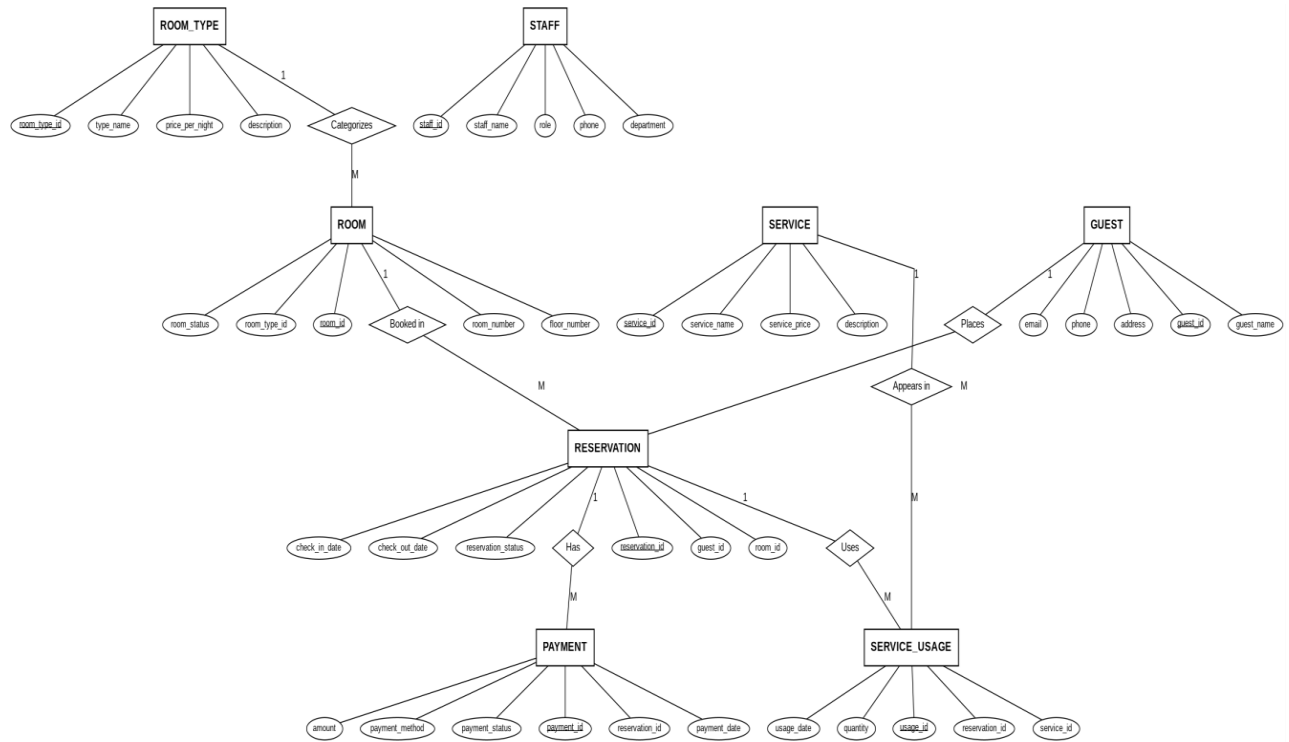
Entity 8: SERVICE_USAGE

Column	Key
usage_id	Primary Key
reservation_id	Foreign Key
service_id	Foreign Key
usage_date	-
quantity	-

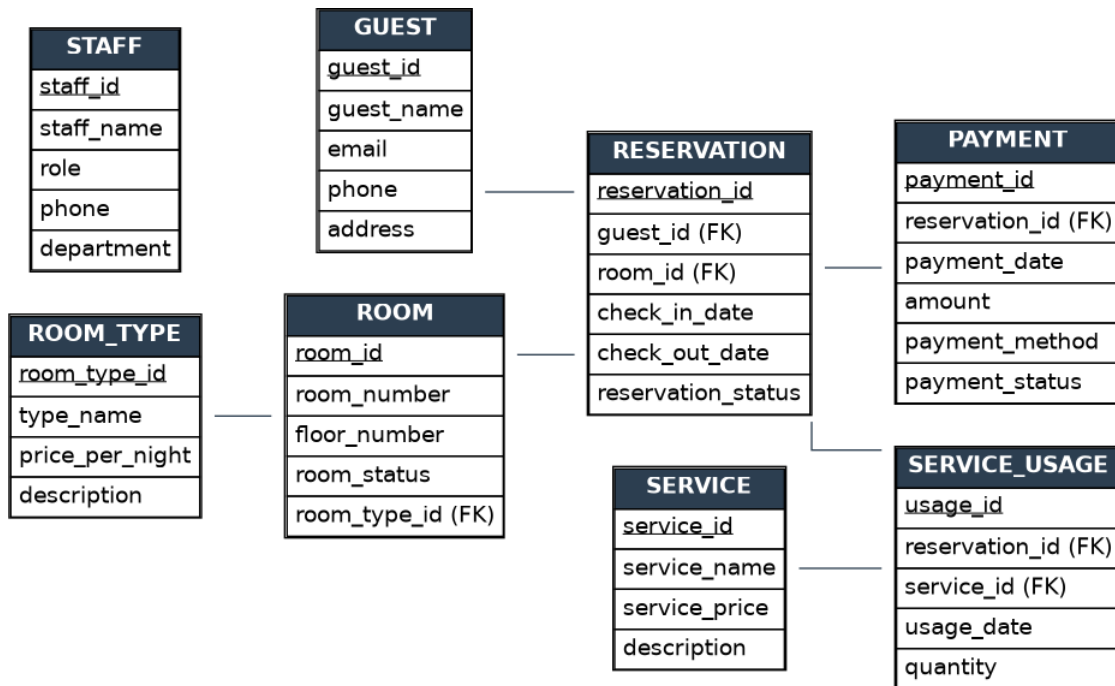
RELATIONSHIPS AND CARDINALITIES

Entity 1	Relationship	Entity 2	Cardinality
GUEST	Places	RESERVATION	1:M
ROOM_TYPE	Categorizes	ROOM	1:M
ROOM	Booked in	RESERVATION	1:M
RESERVATION	Has	PAYMENT	1:M
RESERVATION	Uses	SERVICE_USAGE	1:M
SERVICE	Appears in	SERVICE_USAGE	1:M

ER DIAGRAM



RELATIONAL SCHEMA



PRIMARY KEYS AND FOREIGN KEYS

Table	Primary Key	Foreign Key
GUEST	guest_id	-
ROOM_TYPE	room_type_id	-
ROOM	room_id	room_type_id
STAFF	staff_id	-
RESERVATION	reservation_id	guest_id, room_id
SERVICE	service_id	-
PAYMENT	payment_id	reservation_id
SERVICE_USAGE	usage_id	reservation_id, service_id

RELATIONSHIP EXPLANATION

Guest – Reservation: One guest can place many reservations. Therefore, the relationship is 1:M.

Room Type – Room: One room type can be assigned to many rooms. Therefore, the relationship is 1:M.

Room – Reservation: One room can be booked in many reservations over time. Therefore, the relationship is 1:M.

Reservation – Payment: One reservation can have one or more payment records in this model. Therefore, the relationship is 1:M.

Reservation – Service_Usage: One reservation can include many service-usage records. Therefore, the relationship is 1:M.

Service – Service_Usage: One service can appear in many service-usage records. Therefore, the relationship is 1:M.